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Integrating Artificial Intelligence into Nigeria's Healthcare System

*A Comparative Analysis of Implementation Pathways, Benefits, Risks, and Regulatory
Lessons from Seven Countries*

**RWANDA • KENYA • INDIA • UNITED KINGDOM • CHINA • UNITED STATES •
NIGERIA**

Research & Policy Paper
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Executive Summary

Nigeria faces a critical healthcare gap characterised by high disease burden, workforce shortages, and limited rural access, yet lacks AI-specific health legislation to guide technological integration. This comparative case study analyses AI implementation pathways across seven countries — Rwanda, Kenya, India, the United Kingdom, China, the United States, and Nigeria — to identify transferable regulatory, operational, and clinical lessons for Nigeria.

Using secondary data from peer-reviewed literature, policy documents, and programme evaluation reports, the study examines regulatory frameworks, implementation models, measured health outcomes, and identified risks in each context. Findings indicate that successful AI integration typically requires: (1) clear regulatory classification, (2) national coordination mechanisms, (3) public-private partnerships targeting high-burden diseases, (4) workforce capacity building, and (5) continuous post-market monitoring.

Nigeria's existing Qure.ai tuberculosis screening programme, reported at over 400 portable X-ray units, demonstrates technical feasibility but has not yet achieved national scale. This paper proposes a three-phase roadmap (2026–2035) that combines Rwanda's public-private logistics model, India's infrastructure-first strategy, the United Kingdom's centralised evaluation body, and the United States' risk-based device classification. Projected outcomes — contingent on implementation fidelity and drawn from international benchmarks rather than Nigeria-specific trials — include a 27% reduction in tuberculosis-related adverse outcomes, a 46–51% reduction in postpartum haemorrhage deaths, and 15–40% operational cost savings in comparable health-financing contexts.

This study is intended to inform Nigeria's pending Digital Health Act and National AI Strategy, and to support engagement with development partners and research funders active in African digital health.

Keywords: *artificial intelligence; healthcare; Nigeria; comparative health policy; digital health regulation; LMIC health systems; public-private partnerships*

1. Comprehensive Research Topic

“Integrating Artificial Intelligence into Nigeria's Healthcare System: A Comparative Analysis of Implementation Pathways, Benefits, Risks, and Regulatory Lessons from Seven Countries (Rwanda, Kenya, India, the United Kingdom, China, the United States, and Nigeria)”

2. Research Question

How can Nigeria effectively integrate artificial intelligence into its healthcare system to improve health outcomes, what are the principal benefits and risks of doing so, and what regulatory and implementation lessons can be drawn from comparative case studies of Rwanda, Kenya, India, the United Kingdom, China, and the United States?

3. Rationale

Nigeria faces significant health system challenges, including a high communicable and non-communicable disease burden, workforce shortages, and limited access to quality care, particularly in rural areas. Artificial intelligence has emerged as a transformative technology capable of addressing these challenges through improved diagnostics, expanded access, and operational efficiency. Nigeria, however, currently lacks AI-specific health legislation, and oversight of existing AI-enabled health tools remains fragmented across multiple agencies with weak enforcement mechanisms.

This research draws lessons from seven countries spanning a deliberately diverse range of income levels, regulatory traditions, and implementation models — three African nations (Rwanda, Kenya, Nigeria), two Asian countries (India, China), and two Western countries (the United Kingdom, the United States) — to develop a context-appropriate, phased roadmap for Nigeria. The comparative approach is intended to surface governance principles that are transferable across health systems, rather than to suggest that any single country's model can be adopted wholesale.

4. Methodology

4.1 Design

This is a comparative case study analysis using secondary data drawn from peer-reviewed literature, national policy documents, regulatory databases, and published programme evaluation reports. No primary data collection was undertaken.

4.2 Data Sources

- Nigeria: National Health Act (2014); Nigeria Data Protection Act (2023); draft National AI Strategy; published case material on the Qure.ai tuberculosis screening programme.
- Rwanda: Zipline drone-delivery programme evaluation reports; Rwanda Ministry of Health publications.
- Kenya: Digital Health Act (2023); M-TIBA programme outcome reporting.
- India: Ayushman Bharat Digital Mission (ABDM) documentation; National Health Authority reports.
- United Kingdom: NHS AI Lab evaluation reports; NHS England digital health publications.
- China: National Medical Products Administration (NMPA) guidelines; national AI healthcare strategy documents.
- United States: FDA Software as a Medical Device (SaMD) database; hospital AI-adoption surveys; peer-reviewed outcome studies.

4.3 Analysis Framework

- Regulatory environment — type of AI-specific legislation, medical device classification, and enforcement mechanisms.
- Implementation approach — top-down versus market-driven; public-private partnerships; national coordination.
- Clinical areas — priority diseases or departments where AI is deployed.
- Measured outcomes — quantitative improvements in health outcomes, efficiency, and cost.
- Identified risks — implementation barriers, equity concerns, and unintended consequences.
- Transferability — lessons applicable to Nigeria's institutional and infrastructural context.

4.4 Evidentiary Standard

Each quantitative claim reported in this paper has been reviewed against its underlying source and classified as either independently published (drawn from a named institutional report, regulatory database, or peer-reviewed source) or pending verification (reported in secondary or grey literature but not yet confirmed against a primary source at the time of writing). Claims in the latter category are flagged throughout Section 6 and consolidated in Appendix A, in keeping with standard practice for policy papers intended for external funding or journal submission.

5. AI Typology Framework

A recurring methodological challenge in cross-country AI-health comparisons is that the term “artificial intelligence in healthcare” spans fundamentally different classes of technology, with different evidentiary standards, risk profiles, and regulatory treatment. Comparing a drone logistics

algorithm to a diagnostic imaging model, for instance, conflates distinct intervention types. The typology below classifies each country's primary AI application by function, and is used throughout this paper to frame synthesis around governance principles rather than direct numerical benchmarking across dissimilar technologies.

Country	Primary AI Type	Function	Regulatory Treatment
Rwanda	Logistics AI	Route optimisation for autonomous blood and medical-product delivery	Operational policy, not medical device law
Kenya	Administrative / Financing AI	Automated insurance-claims processing and chronic-disease risk flagging	Data-governance statute (Digital Health Act)
India	Infrastructure + Clinical Decision Support	Identity-linked digital records; triage; TB and retinopathy screening	Infrastructure-first; AI guidelines in development
United Kingdom	Clinical Decision Support + Administrative AI	Stroke-imaging triage; ambient clinical documentation	Centralised evaluation body (NHS AI Lab) + EU AI Act
China	Diagnostic Imaging AI	Medical imaging and primary-care triage at grassroots scale	Uniform Class III (highest-risk) device classification
United States	Diagnostic + Predictive + Administrative AI	Imaging, sepsis/readmission prediction, ambient documentation, billing	Risk-based SaMD framework; 96% Class II
Nigeria	Diagnostic Imaging AI	Portable X-ray-based tuberculosis screening	No AI-specific health legislation

Table 1. AI typology by primary function and regulatory treatment, seven-country comparison.

Three implications follow. First, Nigeria's only mature AI deployment (diagnostic imaging) sits in the same functional category as China's and the United States' most heavily regulated or most rapidly scaled applications — suggesting imaging is a natural first candidate for formal Nigerian device classification. Second, logistics and financing AI (Rwanda, Kenya) have advanced furthest under general operational or data-governance policy rather than dedicated medical-device law, indicating Nigeria could pursue rapid gains in these categories without waiting for comprehensive AI-health legislation. Third, clinical decision support — the most safety-critical category — is, without exception, deployed only within countries that have either a centralised evaluation body (UK) or a risk-tiered device framework (US, China, India in progress), reinforcing that this category should be sequenced after Nigeria's regulatory foundation is in place.

6. Case Study Findings

Each of the following case studies presents the regulatory approach, implementation model, clinical priority areas, measured outcomes, principal risks, and a single transferable lesson for Nigeria. Figures drawn from independently published institutional sources are presented without qualification; figures reported only in secondary or grey literature at the time of writing are marked “pending verification” and are consolidated with source-tracing notes in Appendix A.

6.1 Nigeria (Baseline)

Regulatory Approach: Fragmented oversight across the National Health Act (2014), the Nigeria Data Protection Act (2023), and a draft National AI Strategy; no AI-specific health legislation currently exists.

Implementation: Private sector-led pilots — principally Qure.ai tuberculosis screening — with limited government coordination.

Clinical Areas: Tuberculosis screening (AI-powered portable X-ray), maternal health (digital stethoscopes), and fever-diagnosis expert systems.

Outcomes

- **TB screening scale:** expanded from a single unit to 400+ portable X-ray units nationally *[pending verification]*
- **Diagnostic turnaround:** reduced from days to minutes at point of care *[pending verification]*
- **Access:** improved diagnostic reach into rural and underserved areas

Risks: Poor supporting infrastructure (electricity, connectivity), data-privacy concerns, low AI literacy among clinicians, and regulatory gaps.

Key Lesson: *AI can function in Nigeria even under weak regulation, but will not achieve scale or equity without national coordination.*

6.2 Rwanda — AI-Coordinated Drone Delivery

Regulatory Approach: Public-private partnership under Ministry of Health oversight; no standalone AI law, but integrated into national health-logistics policy.

Implementation: 2016 partnership with Zipline; government-led, structured around performance-based delivery contracts.

Clinical Areas: Maternal health (blood products for postpartum haemorrhage) and paediatric care (vaccines, malaria treatment).

Outcomes

- **Blood delivery time:** reduced from 3–5 hours to 15–49 minutes (approximately 66% faster)
- **Maternal deaths from postpartum haemorrhage:** down 46–51% within one year of deployment
- **Blood product wastage:** down 67%
- **Child mortality from severe malnutrition:** reported down 22% *[pending verification]*

Risks: Donor dependency, high initial capital cost, and the need for continuous regulatory oversight as the logistics network scales.

Key Lesson: Targeted public-private partnerships, tightly scoped to a single high-burden clinical problem, can achieve measurable mortality reductions faster than broader digital-health reforms.

6.3 Kenya — M-TIBA Health Financing

Regulatory Approach: Digital Health Act (2023) provides the data-governance framework; AI is embedded within existing health-financing policy rather than governed by standalone AI legislation.

Implementation: A mobile health-savings wallet with over 4.8 million registered users, layered with AI-driven claims automation rolled out in 2024–2025.

Clinical Areas: Health financing, insurance-claims processing, and chronic-disease management (hypertension, diabetes).

Outcomes

- **Claims approval time:** reduced from multiple days to under 12 hours (approximately 95% faster) *[pending verification]*
- **Healthcare costs to enrolled members:** reported down by up to 15% *[pending verification]*
- **Blood-pressure control among enrolled hypertensive patients:** up from 42% to 50% *[pending verification]*
- **Diabetes control among enrolled diabetic patients:** up from 52% to 74% *[pending verification]*

Risks: Digital divide in remote areas, mobile network coverage gaps, and risk of financial exclusion for populations without smartphone access.

Key Lesson: Enacting data-governance legislation with AI-relevant clauses — even without AI-specific law — can unlock measurable efficiency and cost gains at national scale.

6.4 India — National Digital Health Mission

Regulatory Approach: Centralised digital infrastructure through the Ayushman Bharat Digital Mission (ABDM); dedicated AI guidelines under development for 2025–2026.

Implementation: A national health-identity system linking 104 crore (1.04 billion) records to 93 crore (930 million) ABHA IDs, with AI embedded in tuberculosis screening, diabetic retinopathy detection, and e-prescription workflows.

Clinical Areas: Primary-care triage, tuberculosis elimination, diabetic retinopathy screening, and e-prescriptions.

Outcomes

- **TB outcomes:** 27% reduction in adverse outcomes; 12–16% increase in case detection *[pending verification]*
- **Clinic efficiency:** over 1 million patients triaged and 1.7 million records summarised via AI documentation tools *[pending verification]*
- **Diabetic retinopathy screening:** 7,100 patients screened across 38 facilities *[pending verification]*
- **Digital identity infrastructure:** 104 crore records linked to 93 crore ABHA IDs

Risks: Data-privacy concerns at national scale, interoperability gaps between state health systems, and clinician workforce-training needs.

Key Lesson: *Building national digital health infrastructure first — before layering AI tools on top of it — enables large-scale impact once AI is introduced.*

6.5 United Kingdom — NHS AI Lab

Regulatory Approach: The NHS AI Lab (launched 2019) alongside EU AI Act compliance obligations; centralised funding, standardised evaluation, and national procurement.

Implementation: National rollout across 107 stroke centres, 1,600+ general-practice sites, and 200+ hospital trusts, with AI integrated directly into existing electronic health records.

Clinical Areas: Stroke imaging, administrative automation (clinical note-taking, discharge summaries), and diagnostic support for lung cancer and diabetic retinopathy.

Outcomes

- **Stroke treatment time:** reduced 44% (from 140 to 79 minutes); disability-free discharge up 32 percentage points (16% to 48%)
- **Administrative efficiency:** 47 minutes saved per clinician per shift; approximately 400,000 hours saved per month nationwide
- **Direct cost savings:** £44 million saved across 150,000 patients against a tool cost of £1.25 million
- **Missed appointments:** reported down 30% (377 prevented over six months), saving an estimated £27.5 million per year *[pending verification]*

- **Clinician AI training:** only 26% of clinicians reported receiving in-service AI training [*pending verification*]

Risks: High upfront implementation costs, uneven workforce readiness, interoperability challenges across legacy NHS systems, and rural-urban disparities in rollout.

Key Lesson: *A national AI evaluation body with standardised procurement enables measurable improvements at scale, but workforce training must keep pace with deployment.*

6.6 China — NMPA Class III and National Strategy

Regulatory Approach: All AI medical software is classified as Class III (highest risk), requiring stringent National Medical Products Administration (NMPA) review; implementation follows a top-down national strategy with cross-ministerial coordination.

Implementation: Five central ministries jointly promote AI integration, with a 2027 target for high-quality shared datasets and a 2030 target for AI deployment across all primary-care facilities.

Clinical Areas: Medical imaging (67% of NMPA-approved AI devices), primary-care triage, and smart health platforms.

Outcomes

- **NMPA-approved AI devices:** 132 approved by 2023, compared with over 1,000 FDA-approved devices in the United States over the same period
- **Deployment pace:** rapid nationwide extension to grassroots clinics under central coordination
- **Affordability:** government-regulated pricing keeps approved AI tools accessible to public hospitals

Risks: Regulatory stringency that slows innovation relative to risk-tiered systems, data-quality concerns, and algorithmic bias arising from urban-trained models applied in rural settings.

Key Lesson: *Government-led coordination with standardised datasets accelerates national scale, but uniform high-risk classification for all AI trades innovation speed for caution — a trade-off Nigeria should make deliberately rather than by default.*

6.7 United States — FDA SaMD and Market-Driven Adoption

Regulatory Approach: FDA Software-as-a-Medical-Device (SaMD) framework, under which 96% of authorised AI devices are classified Class II (moderate risk), enabling materially faster approval than China's uniform Class III standard.

Implementation: Market-driven innovation led by private companies, technology firms, and academic medical centres; adoption decisions are made at the individual hospital level, with no national mandate.

Clinical Areas: Imaging and radiology (used by 90% of organisations surveyed), ambient clinical documentation, predictive analytics for sepsis and readmission risk, and administrative automation.

Outcomes

- **FDA-approved AI/ML-enabled devices:** over 1,000 by 2025, compared with 132 in China
- **Physician AI adoption:** reported at 66% by 2025, up from 18.7% in 2022 *[pending verification]*
- **Clinical impact:** predictive AI tools associated with reduced pneumonia mortality, fewer hospital-acquired conditions, and shorter length of stay
- **Administrative impact:** ambient AI documentation saves an estimated 47 minutes per clinician shift; billing automation reduces administrative burden
- **Adoption disparity:** only 3.8% of hospitals classified as “high adopters” in a 2022 benchmark survey *[pending verification]*

Risks: Significant adoption disparities between well-resourced and smaller hospitals, workflow-integration challenges, clinician burnout linked to tool proliferation, and cost barriers for under-resourced facilities.

Key Lesson: *Market-driven innovation produces the largest device catalogue and the fastest approvals, but without national coordination it concentrates benefit in already well-resourced institutions — a risk directly relevant to Nigeria's urban-rural divide.*

7. Comparative Analysis Table

Country	Regulatory Model	Approval Speed	Scale	Measured Outcomes	Key Risk
Rwanda	Public-private partnership	Fast (MoH-led)	National (all public hospitals)	Blood delivery –66%; PPH deaths –46–51%; blood wastage –67%	Donor dependency
Kenya	Digital Health Act (2023)	Moderate (legislative)	4.8M+ users	Claims time –95%*; costs –15%*; BP control +8pp*; diabetes control +22pp*	Digital divide
India	National Digital Health Mission	Slow (infrastructure-first)	1.04bn records	TB adverse outcomes –27%*; 1M+ triaged; 7,100 retinopathy screenings*	Interoperability gaps
United Kingdom	NHS AI Lab + EU AI Act	Moderate (standardised evaluation)	107 stroke centres; 1,600+ GP practices	Stroke treatment –44%; admin savings 400K hrs/mo; £44M cost savings	High upfront cost
China	NMPA Class III (all AI)	Slow (stringent review)	132 approved devices; nationwide by 2030	Rapid grassroots deployment; regulated affordable pricing	Innovation slowdown
United States	FDA SaMD (96% Class II)	Fast (market-driven)	1,000+ approved devices; 66% physician adoption*	Pneumonia mortality down; HAC down; 47 min/shift saved	Adoption disparities
Nigeria	Fragmented (no AI-)	Variable (pilot-dependent)	400+ TB screening units*	TB turnaround days→minutes*;	Lack of scale/equity

Country	Regulatory Model	Approval Speed	Scale	Measured Outcomes	Key Risk
	specific law)			rural access improved	

Table 2. Cross-country comparison of regulatory model, approval speed, scale, and outcomes. Asterisk () denotes figures pending independent verification — see Appendix A. “pp” denotes percentage points.*

8. Synthesis: Key Patterns Across Countries

8.1 What Works

- National coordination accelerates scale. India, the United Kingdom, and China each demonstrate that a centralised digital infrastructure or evaluation body enables nationwide deployment that fragmented, pilot-by-pilot approaches cannot match.
- Public-private partnerships reduce cost and time to impact. Rwanda's Zipline model and Kenya's M-TIBA programme show that tightly scoped, government-led partnerships can achieve measurable outcomes faster than either pure state provision or pure market approaches.
- Starting with high-burden diseases concentrates impact. Every country reviewed prioritised AI deployment for tuberculosis, maternal health, stroke, or diabetes — conditions already carrying the highest disease burden in each context.
- Regulatory clarity builds clinician and patient trust. Countries with clear AI device classification (FDA Class II, NMPA Class III, the EU AI Act) report higher clinician confidence and smoother patient acceptance than jurisdictions without such clarity.

8.2 What Does Not Work

- Fragmented oversight slows adoption. Nigeria's current patchwork of general health, data-protection, and draft AI policy creates persistent uncertainty for both developers and hospital administrators.
- Market forces alone create disparities. The United States shows the largest device catalogue of any country reviewed, yet only 3.8% of hospitals qualify as “high adopters” — rural and under-resourced facilities are structurally left behind absent coordinating policy.
- Uniform high-risk classification slows beneficial innovation. China's blanket Class III requirement for all AI software results in materially slower approval than the FDA's risk-tiered approach, without a demonstrated corresponding safety advantage in the cases reviewed.

- Pilots without a scaling mechanism stall. Nigeria's Qure.ai programme is a genuine technical success, but in the absence of a national scaling mechanism it has not translated into national tuberculosis-screening coverage.

9. Limitations

This study has several limitations that bear on how its findings should be interpreted and used.

Publication and reporting bias

The secondary sources reviewed disproportionately represent successful AI deployments. Programmes that underperformed, were discontinued, or produced null results are underrepresented in published literature and government reporting. Rwanda's Zipline outcomes, for example, are drawn primarily from Ministry of Health and partner evaluations, which may understate periods of operational difficulty.

Cross-country comparability

The seven cases span fundamentally different AI typologies — logistics (Rwanda), health financing (Kenya), diagnostic imaging (Nigeria, China, the United States), clinical decision support (India, the United Kingdom), and administrative automation (the United Kingdom, the United States). Direct numerical comparison of outcomes (for instance, “66% faster delivery” against “44% faster stroke treatment”) conflates distinct intervention types, health-system structures, and baseline conditions. This paper therefore emphasises governance principles and implementation patterns — formalised in the typology in Section 5 — over direct outcome benchmarking.

Causal inference

Reported outcomes are associations observed in real-world programme data, not randomised controlled trial results. Confounding factors — concurrent workforce training, infrastructure investment, broader health-system reform, and secular disease trends — may contribute to observed improvements. Findings throughout this paper are framed as “associated with” rather than “caused by,” and should be read as real-world evidence rather than attributable causal effect.

Nigeria-specific data gaps

Several Nigeria-specific figures, including the precise number of currently operational Qure.ai portable X-ray units and exact diagnostic turnaround metrics, are drawn from secondary reporting and implementer communications rather than a verified primary source. These figures are flagged throughout Section 6 and consolidated in Appendix A, and require confirmation with implementing partners — Qure.ai, the Institute of Human Virology Nigeria (IHVN), and the Federal Ministry of Health — before formal publication or citation in funding proposals.

Temporal variation

Case studies span implementation periods from 2016 (Rwanda) to 2025–2026 (Kenya's AI claims automation; India's AI guidelines). Health-system context, technology maturity, and disease epidemiology have shifted materially over this period, which limits strict temporal comparability across cases.

Scope constraints

Seven countries, while deliberately diverse, cannot capture the full spectrum of AI-health governance models. Notably absent are perspectives from Latin America (for example, Brazil's SUS digital health programme) and Southeast Asia (for example, Singapore's HealthTech regulatory sandbox), which may offer additional transferable lessons for future research.

10. Recommendations for Nigeria

Drawing on the comparative evidence above, Nigeria is best served by a hybrid model that combines: Rwanda and Kenya's public-private partnership approach for high-burden diseases; India's infrastructure-first strategy for national health identity and records; the United Kingdom's centralised evaluation body as a model for a Nigerian equivalent; and the United States' risk-based device classification, in preference to a uniform high-risk standard such as China's.

Phase 1 (2026–2027): Enabling Environment

- Enact a Digital Health Act with AI-specific clauses covering data governance, accountability, and interoperability.
- Establish a National Health AI Certification Body under the Federal Ministry of Health, structured on the NHS AI Lab model.
- Scale the Qure.ai tuberculosis-screening programme nationally from 400+ to a target of 2,000+ units, building on a proven operational model.

Phase 2 (2028–2030): Infrastructure and Standards

- Build a National Health Data Repository with standardised datasets, drawing on India's ABDM implementation experience.
- Launch an AI Workforce Training Programme for clinicians, informed by the UK's identified 26% in-service training gap.
- Develop a Risk-Based AI Classification Framework — Class II for low-risk applications, Class III reserved for high-risk clinical decision support — rather than defaulting to uniform high-risk treatment.

Phase 3 (2030–2035): National Scale

- Deploy AI-powered maternal-health logistics through a public-private partnership modelled on Rwanda's Zipline programme.
- Integrate AI-assisted claims processing into the National Health Insurance Authority, drawing on Kenya's M-TIBA experience.
- Establish a Post-Market AI Monitoring System for continuous real-world performance tracking across all certified tools.

11. Expected Outcomes

If implemented with fidelity, this roadmap could plausibly achieve, based on the international benchmarks reviewed in this paper:

- Tuberculosis detection: a 27% reduction in adverse outcomes (India benchmark).
- Maternal health: a 46–51% reduction in postpartum haemorrhage deaths (Rwanda benchmark).
- Administrative efficiency: approximately 47 minutes saved per clinician per shift (UK/US benchmark).
- Cost savings: 15–40% operational cost reduction in comparable financing contexts (Kenya/UK benchmarks).
- Rural access: up to 66% faster diagnostic or logistics turnaround (Rwanda/India benchmarks).

These are projected outcomes based on international benchmarks and are not guaranteed. Actual results in Nigeria will depend on implementation fidelity, funding continuity, workforce readiness, and contextual factors specific to Nigeria's health system, and should be treated as planning targets rather than forecasts.

12. Conclusion

This comparative analysis demonstrates that AI integration in healthcare is not a binary choice between “regulated” and “unregulated” models, but a spectrum of approaches that can be adapted to Nigeria's specific institutional and infrastructural context. Evidence from seven countries indicates that successful AI integration typically requires clear regulatory frameworks, national coordination mechanisms, targeted public-private partnerships for priority clinical areas, sustained workforce capacity building, and continuous post-market monitoring and evaluation.

Nigeria's current fragmented approach constrains AI's potential contribution to health system strengthening. A deliberate, phased strategy — grounded in the international lessons set out in this paper, and sequenced to match Nigeria's present regulatory and infrastructural readiness — can

accelerate progress toward universal health coverage while managing the risks that have accompanied faster-moving deployments elsewhere.

13. References

Citation style: Vancouver (numbered). Entries marked “to be confirmed” correspond to the pending-verification items listed in Appendix A and should be finalised before external submission.

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Appendix A: Quantitative Claims Verification Log

This paper contains 23 discrete quantitative claims drawn from secondary literature. Of these, 11 are traceable to an independently published institutional source and are classified “Published” below; 12 require further source confirmation before the paper is cited externally or submitted for publication, and are classified “Pending.” This log is provided for editorial transparency and should be treated as a working document, updated as sources are confirmed.

#	Claim as stated in paper	Section	Status	Action required
1	400+ portable X-ray units (Nigeria, Qure.ai)	§6.1	Pending	Contact Qure.ai / IHVN for exact operational unit count as of August 2026.
2	TB diagnostic turnaround “days to minutes”	§6.1	Pending	Request specific median turnaround data from implementing partner.
3	Blood delivery 3–5 hrs → 15–49 min (Rwanda)	§6.2	Published	Zipline / Rwanda MoH evaluation report; cite publication year.
4	Maternal PPH deaths down 46–51%	§6.2	Published	Rwanda MoH impact evaluation; verify exact time window.
5	Blood wastage down 67%	§6.2	Published	Same source as above.
6	Child mortality (severe malnutrition) down 22%	§6.2	Pending	Identify the specific evaluation report and time window.
7	4.8M+ M-TIBA users (Kenya)	§6.3	Published	Verify against M-TIBA / Safaricom 2025–26 annual report.
8	Claims approval “days to under 12 hours”	§6.3	Pending	Confirm rollout-phase (2024–25) data availability.
9	Healthcare costs down up to 15%	§6.3	Pending	Identify source publication.
10	BP control 42%→50%; diabetes 52%→74%	§6.3	Pending	Identify peer-reviewed or programme-evaluation citation.
11	1.04bn records / 930M ABHA IDs (India)	§6.4	Published	Cite ABDM / NHA dashboard; note access date.
12	TB adverse outcomes down 27%; detection up 12–16%	§6.4	Pending	Identify specific NHA or WHO India report.
13	1M+ triaged; 1.7M records summarised (Eka Doc)	§6.4	Pending	Verify with Eka Care / NHA publications.
14	7,100 retinopathy screenings, 38 facilities	§6.4	Pending	Identify pilot report or peer-reviewed paper.
15	Stroke treatment 140→79 min; discharge 16%→48%	§6.5	Published	NHS AI Lab / NHS England evaluation report.
16	47 min/shift saved; 400,000 hrs/month	§6.5	Published	NHS England digital health publications.
17	£44M saved across 150,000 patients	§6.5	Published	NHS AI Lab report; verify exact publication year.
18	Missed appointments down 30%; £27.5M/yr saved	§6.5	Pending	Identify specific trust-level or national report.
19	Only 26% receive in-service AI training (UK)	§6.5	Pending	Cite survey / report.
20	132 NMPA-approved AI devices by 2023 (China)	§6.6	Published	NMPA public database; note access date.

#	Claim as stated in paper	Section	Status	Action required
21	1,000+ FDA-approved AI devices by 2025 (USA)	§6.7	Published	FDA SaMD database; note access date.
22	66% of physicians using AI by 2025 (up from 18.7%)	§6.7	Pending	Identify specific survey (AMA, Medscape, or KLAS).
23	Only 3.8% of hospitals “high adopters” in 2022	§6.7	Pending	Identify survey / report.

Table A1. Full verification status of all quantitative claims in this paper, by section.

Summary: 23 quantitative claims identified; 11 verified against a published institutional source; 12 pending verification. Recommended action prior to external circulation: contact implementing partners directly for Nigeria-specific figures (claims 1–2), and identify primary survey or programme-evaluation citations for the remaining pending items. Where an exact figure cannot be confirmed by the submission deadline, use hedging language (“approximately,” “reported as,” “as of [date]”) rather than presenting the figure as settled fact.

Appendix B: Peer Review Readiness & Publication Strategy

This appendix summarises the paper's readiness for external submission — to academic journals, development-partner review, or funder due diligence — and sets out concrete steps to close the remaining gaps identified in Appendix A.

B.1 Strengths for External Review

- A seven-country comparative framework spanning African, Asian, and Western regulatory models, which is not, to the authors' knowledge, replicated elsewhere in the current literature on Nigerian digital health policy.
- An explicit AI typology (Section 5) that separates logistics, financing, diagnostic, and clinical-decision-support applications — addressing the most common reviewer objection to cross-country AI-health comparisons, namely that they conflate dissimilar technologies.
- A dedicated limitations section (Section 9) addressing publication bias, causal inference, and cross-country comparability in line with standard methodological expectations for comparative policy research.
- Full source-transparency via the verification log in Appendix A, distinguishing published findings from claims still requiring primary confirmation.
- A phased, time-bound roadmap (Section 10) rather than a purely descriptive comparison, giving the paper direct relevance to Nigeria's pending Digital Health Act and National AI Strategy.

B.2 Before External Submission

- Resolve the 12 pending items in Appendix A, prioritising the two Nigeria-specific figures (Qure.ai unit count and diagnostic turnaround), since these most directly affect the paper's core policy argument.
- Confirm final publication status of Nigeria's draft National AI Strategy (reference 3) and update the citation accordingly.
- Consider commissioning a brief independent methodological review from a health-policy researcher outside LRI prior to journal submission, standard practice for comparative policy papers at this stage.
- If the paper exceeds a target journal's word count once pending items are resolved, consider splitting into a shorter policy brief (for funder and government audiences) and a longer academic version (for journal submission).

B.3 Potential Outlets

Tier	Suitable outlets
Tier 1 — high impact	BMJ Global Health; The Lancet Digital Health; Health Policy and Planning.
Tier 2 — strong scope fit	Journal of Global Health; Digital Health (SAGE); AI & Society.
Tier 3 — regional	African Journal of Primary Health Care & Family Medicine; Nigerian Medical Journal.

Tier	Suitable outlets
Non-journal	Direct submission to Federal Ministry of Health, NITDA/NCAIR, and funders such as Coefficient Giving and the Wellcome Trust as a standalone policy paper.

Table B1. Potential publication and dissemination outlets by tier.

Tier 1 journals are competitive for a secondary-data comparative paper of this kind, but the seven-country scope and explicit typology framework are genuine differentiators. Tier 2 and Tier 3 outlets, along with direct submission to government and funder audiences, represent the most probable near-term paths to impact and should be pursued in parallel rather than sequentially.